

ZyraClothing — Full Project Summary

Live Deployment

<https://ecommerce-c8kwx1jmr-lahiru-a01s-projects.vercel.app>

What We Built

A **fully functional e-commerce web application** called **ZyraClothing** with real product data, user authentication, shopping cart, protected routes, and live deployment.

Complete Tech Stack

Frontend

Technology	Version	Purpose
React	v18	UI component framework
Vite	v8	Build tool & dev server
React Router v6	v6	Page navigation & routing
Tailwind CSS	v3	Utility-first styling
Lucide React	latest	Icons (cart, login, logout)
React Hot Toast	latest	Success/error notifications

State Management

Technology	Purpose
Zustand	Global state management
zustand/persist	Persist cart & auth to localStorage

API & Data

Technology	Purpose
Axios	HTTP requests
FakeStoreAPI	Free REST API for products & auth

Version Control & Deployment

Technology Purpose

Git	Version control
GitHub	Remote repository hosting
Vercel	Live deployment & hosting

Project Structure

ecommerce-app/

```
├── public/
├── src/
│   ├── api/
│   │   └── products.js    # All API calls
│   ├── components/
│   │   ├── Navbar.jsx    # Navigation bar
│   │   ├── ProductCard.jsx # Product display card
│   │   ├── CartItem.jsx   # Cart row item
│   │   └── ProtectedRoute.jsx # Auth route guard
│   ├── pages/
│   │   ├── Home.jsx      # Product listing page
│   │   ├── ProductDetail.jsx # Single product page
│   │   ├── Cart.jsx       # Shopping cart page
│   │   ├── Login.jsx      # Authentication page
│   │   └── NotFound.jsx    # 404 page
│   ├── store/
│   │   ├── cartStore.js   # Cart state (Zustand)
│   │   └── authStore.js   # Auth state (Zustand)
│   ├── App.jsx            # Routes & layout
│   └── main.jsx           # App entry point
```

└─ index.css	# Tailwind imports
└─ index.html	# App shell + title
└─ vercel.json	# SPA routing fix
└─ tailwind.config.js	# Tailwind config
└─ postcss.config.js	# PostCSS config
└─ vite.config.js	# Vite config

Features Built

Shopping

- Product grid listing with real images & prices
- Category filter (All, Electronics, Jewellery, Men's, Women's)
- Product detail page with full description & rating
- Add to cart from both listing and detail page
- Cart badge count in navbar (live updates)

Authentication

- Login with JWT token via FakeStoreAPI
- Token persisted in localStorage via Zustand
- Protected /cart route — redirects to login if not logged in
- Logout clears token and redirects home
- Hi, username display in navbar when logged in

Cart

- Add items with quantity tracking
- Increase / decrease quantity
- Remove individual items
- Clear entire cart
- Real-time order total calculation
- Persisted in localStorage (survives page refresh)

Routing

Route	Page	Protected
/	Home — product listing	✗ Public
/product/:id	Product detail	✗ Public
/login	Login form	✗ Public
/cart	Shopping cart	✓ Login required
/*	404 Not Found	✗ Public

App Data Flow

FakeStoreAPI



Axios (api/products.js)



React Pages (Home, Detail, Login)



Zustand Stores (cartStore, authStore)



Components (Navbar, ProductCard, CartItem)



User Interface (Tailwind CSS)



Deployed on Vercel ← GitHub ← Git

Deployment Pipeline

You write code



git add . && git commit



git push → GitHub repo



Vercel detects push automatically



Vite builds → dist/ folder



Live on Vercel URL in ~30 seconds 

Project Stats

Metric	Count
Total Files Created	11
Pages Built	5
Components Built	4
Zustand Stores	2
API Functions	5
Routes Configured	5
npm Packages Installed	8
Phases Completed	6

Skills You Learned

-  React component architecture
-  React Router v6 (nested, protected routes)
-  Zustand state management
-  Persist state to localStorage
-  REST API integration with Axios
-  Tailwind CSS utility styling
-  JWT-based authentication flow

- ✓ Protected route pattern
 - ✓ Git version control workflow
 - ✓ GitHub remote repository
 - ✓ Vercel deployment & CI/CD
 - ✓ SPA routing fix (vercel.json)
 - ✓ Environment-based builds (Vite)
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Next Steps to Go Professional

Priority	Feature	Tech
 High	Real database	Supabase
 High	Google OAuth	Supabase Auth
 High	Search + filters	React state
 Medium	Stripe payments	Stripe API
 Medium	Dark mode	Tailwind dark:
 Medium	Wishlist	Zustand store
 Low	PWA support	vite-plugin-pwa
 Low	Unit tests	Vitest
 Low	TypeScript	TS migration